The second law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain the direction of heat transfer and the limits of energy conversion.

SUCCESS CRITERIA - I can...

1. Predict spontaneous net heat flow.
2. Explain why moving heat from cold to hot needs an additional process.
3. Distinguish energy conservation from process direction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Samples at **13°C** and **55°C** are mixed with no energy input or phase change. A student reports **67°C** at equilibrium. Is that possible? Explain.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $Q_{hot} = Q_{cold} + W_{in}$.
Refrigerator cycle: heat amounts are positive magnitudes. Work input moves heat from cold to hot; spontaneous flow is hot to cold.

Worked reasoning: Hot and cold blocks in contact exchange energy until thermal equilibrium. Total energy can be conserved while the temperature difference and ability to drive a heat engine decrease.

Your turn: Draw before-and-after temperature states and the spontaneous heat-flow direction.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A **50°C** block touches a **10°C** block. **Draw net heat flow** and **predict their trend toward equilibrium**.

50°C

10°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

During one modeled refrigerator cycle, **20 J** is removed from the cold compartment and **5 J** of electrical work enters.
1. Draw and **label the energy flows**.
2. **How much energy** must be released to the room?

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A proposed machine takes heat from one reservoir and turns all of it into work in a repeating cycle. What thermodynamic limitation is missing?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An energy account begins with **24 J**. It ends with **16 J** motion energy and **4 J** thermal energy. **Find the unaccounted energy** and suggest a next check. **Show the energy changes** and transfers; **label units**.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Does a refrigerator disprove hot-to-cold spontaneous flow? **Explain the role of work** input.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A passive cooling pack and a powered refrigerator both cool objects.
1. **Draw the energy transfers** for each.
2. **Explain why moving energy** from a colder compartment to a warmer room requires an input in the refrigerator case.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3